IBM

(FOR INTRACOMPANY CORRESPONDENCE)

TO—OFFICE AND OR DEPARTMENT WHQ ATTENTION Mr. D. W. Pendery FROM District 7 Office

SUBJECT AND OR REFERENCE Bigger and Better Machines DATE 7/9/56

I made a call to Arnold Research Center at Tallahoma, Tennessee. As you know, they have the largest wind tunnel facilities in the world there or at least, they are constructing them. While there we had a discussion with Dr. Grapau. During this discussion we talked about digital models and simulation. As a result of some of the thinking done on that occasion, I'd like to make the following comments:

I honestly believe that in the foreseeable future, five to ten years away, we can expect to replace wind tunnels to a greater degree than is presently practical by using digital models. However, let's take a quick look at whats required in terms of speeds and storage. If we were to try to determine the air flow over a wing of special shape, we could very easily get into a three dimensional volume that would require a hundred sigments in each of three principle axes. For each of these sigments, we would probably have to retain information on temperature, pressure and a velocity vector. Additional information might be required but this would be a minimum. This implies a storage on the order of magnitude of five million words and dividing the axis into a hundred equal parts is still a very rough model. If we were to divide each axis into a thousand parts, we would have over five billion words of storage required.

In order to do the necessary computation with machines comparable to todays equipment, we would find ourselves hopelessly lost. Consider a case where we were to have a five billion word to be processed with any reasonable amount of arithemetic and all we could easily wind up with is an execution of one hundred to two hundred characters - two hundred instructions being executed per word, or in short, one thrillion instructions would have to be executed. On any reasonably current basis, this is an impossibility.

Despite the challenges of this type of storage and computing speed indicate this is still an area in which digital methods may have avantually a great deal to offer. It opens to conjecture a number of areas, the first is types of storage. It seems impossible to comprehend at the moment a storage that would perhaps have fifty billion or a hundred billion bits in which each piece of each bit storage unit would have to be processed by hand. Storages of this magnitude are going to have to be assembled in some other fashion, for instance, in a fashion comparable to the growing of crystals. In terms of computing speed, speed is not the only answer. For instance, in digital simulation of air flow we essentially want to ply conservation theroms to entire mass of the air, more or less simultaneously.

The success of such techniques will be achieved only through the derivation of a new type of computing unit that essentially has the ability to apply a physical rule to a large number of positions of storage. It further suggests thathe positions of storage must be addressable in such a way that is it possible to call forth essentially with one command all of the cells storing

information about volumns which surround a given volume element.

As I look at this problem, it does not strike me that this is anything which projects stretch will do much to solve. It seems to me to be a completely new generation of machines and one in which big dividends eventually could be reaped which is sometime in the future.

Again, you comments are invited, Don.

J. L. Hawk

JLH/c